

Ishaan Vohra

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Massachusetts Institute of Technology

Cambridge, MA | Expected May 2028

BS candidate in Artificial Intelligence and Decision Making; GPA: 5.0/5.0

Coursework: Machine Learning, Controls, Numerical Methods for Linear Algebra, Algorithms, Lagrangian and Hamiltonian Mechanics, Statistical Inference, Circuits and Electronics, Linear Algebra and Optimization, Differential Equations

Societies: *Co-founder, Co-head* – MIT Undergraduate Robotics Reading Group • *Vice President* – MIT Sloan Robotics Club • *Vice President, Summer House Manager* – MIT Theta Xi Fraternity • *Executive Leadership Team* – MIT Music Production Collaborative

WORK EXPERIENCE

MIT Computer Science and AI Laboratory (CSAIL)

Cambridge, MA

Undergraduate Researcher

Torralba Lab (Prof. Antonio Torralba, Prof. Yilun Du)

May 2026 - present

- Joint project with Harvard Kempner Institute. Multimodal video world models for robotics; visuotactile sensing and force prediction.

Distributed Robotics Lab (Prof. Daniela Rus)

Sep 2025 - present

- Leading project on vision-language-action (VLA) model fine-tuning in simulation; developing novel algorithms for gradient-based optimization of domain randomization parameters.
- Building custom differentiable simulation pipelines; deploying fine-tuned models on robot manipulation hardware (Franka FR3 arm).

Fortyfive Labs

Cambridge, MA | Jan 2026 - Feb 2026

Robot Learning Engineering Intern

- Led development of end-to-end 3D asset generation pipelines to support enterprise robotics clients; generating photorealistic and physically-realistic robot simulation environments in MuJoCo which were demonstrated to investors.
- Built and teleoperated VR-based contact-rich manipulation simulations for clients demos (including *Cobot* and *Pickle Robot*) using robot assets and 3D reconstructed scenes from pipeline.

Quantum Photonics and AI Group, MIT Research Laboratory of Electronics

Cambridge, MA | Mar 2025 - Oct 2025

Undergraduate Researcher

- Built closed-loop, parallelized holographic laser system to correct resonant wavelengths of silicon microring resonators in photonic integrated circuits (PICs) for use in optical neural networks (ONNs).

Fermi National Accelerator Laboratory (Fermilab)

Batavia, IL | Jun 2023 - Jun 2024

Computational Particle Physics Researcher

- Implemented electromagnetic scattering models in the GENIE Event Generator based on quantum Monte Carlo methods.
- Designed and implemented neutrino flux simulations for Fermilab's Deep Underground Neutrino Experiment (DUNE); inducted as a DUNE Collaborator; [journal publication pending](#).

European Organization for Nuclear Research (CERN)

Geneva, Switzerland | Nov 2021 - Apr 2024

Founder and Team Lead, Global Winning Beamline Competition Team

- Led design and experimentation for [world-first](#) radially and rotationally adjustable magnetic Halbach array in CERN's T10 beamline.
- [Lead author](#) of publication presented at *12th Beam Telescopes and Test Beams Workshop (BTTB)*.

University of Liverpool, Department of Physics and Astronomy

Liverpool, United Kingdom | May 2022 - Dec 2022

Honorary Research Fellow, Computational Particle Physics

- Developed neutrino interaction models with GENIE Event Generator; created *NuSmear*, a software package for fast simulation of detector smearing; accepted into GENIE contribution database, accessible to physicists worldwide.
- [Sole author](#) of publication presented at *14th Conference on the Intersections of Particle and Nuclear Physics (CIPANP)*.

SKILLS

Programming Languages: Python, C/C++, Julia, Bash/Shell, Lisp, Wolfram Language

ML: PyTorch, NumPy, Pandas, Matplotlib, HuggingFace

Robotics/Simulation: MuJoCo, NVIDIA Isaac Sim, ANSYS, COMSOL, SolidWorks, LTspice, Tidy3D, Geant4

Tools/Platforms: Git/GitHub, Docker, Linux, CI/CD, AWS, HPC, Jupyter, ROOT, Arduino

AWARDS AND COMPETITIONS

2024: **Founded and led 1st place team** | CERN Beamline Competition – 1st out of >2500 students globally

2023, 2024: **2nd place** | US Invitational Young Physicists' Tournament

2021, 2023: **Finalist** | International Songwriting Competition

2023: **Finalist** | USA Songwriting Competition

2022: **Silver in Physics** | North American Finals, S.T. Yau Science Award